Abdulhamit Güngören

Istanbul, TR | +90 553 080 1982 | Portfolio
abdulhamitgungoren@gmail.com | LinkedIn | Github

PROFESSIONAL PROFILE

Innovative, analytical, and experienced embedded software & AI engineer/architect with a long history of product delivery, software development, creating real-time test and verification platforms and safety-critical embedded solutions. Proven success designing and building system integrations that scale and delivering 85% faster QA and reduced development cycle times. Adept at collaborating with product managers to align feature work with business requirements. Flexible regarding relocation and open to pursuing security clearance.

TECHNICAL SKILLS

C/C++ Programming 17/20, C#, Python 3, .NET 6, Qt 6/QML, CUDA, ROS 2, DDS | ARM Cortex-M, STM32, ESP32, VxWorks, FreeRTOS, RT-Linux | UART, I2C, SPI, RS232/422/485, CAN bus, TCP/UDP | Visual Studio, Qt Creator, CMake, Git/GitLab CI, SVN, Jira, UiPath REFramework | Agile/Scrum, ISO 9001/27001, UML

PROFILE OF SKILLS

Software Development/Design	Algorithms	Software Architecture
Troubleshooting & Testing	Computer Architecture	Design Patterns
Communication	Embedded Systems	Best Practice
Problem Solving	Debugging	Integration

PROFESSIONAL EXPERIENCE

Specialist Software Engineer | Beko – Koç Holding | Tekirdag, TR | 2025 – Present

- Effectively reduce/eliminate flaky interfaces by designing/developing reusable UART/SPI/TCP libraries for hardware-in-loop tests.
- Build and develop personal R&D prototypes in RF signal processing and embedded AI (a passive radar jammer/spoof detector using HackRF One and TinyML).

Key Achievements:

- Saved 1700 hours/month and reduced test cycles by 85% by architecting TVPulse, an IR-simulated Android-TV farm (120 sets).
- Reduced overall development time by 80% and new-toll incubation from weeks to less than 48 hours after creating CoderAI, an internal multi-model code-generation agent that is being considered for enterprise-wide rollout.
- Increased sprint throughput by 25% by integrating GitLab CI/CD with Scrum boards.
- Decreased manual Android-TV regression cycles by ~30% and prototype software build times by ~50% by developing/implementing multiple AI-assisted test-automation tools.
- Designed and introduced “TVPulse” and “TVBenchmark,” platforms, resulting in standardised automated testing and comparative analysis for new TV models.

Software Engineer | Beko – Koç Holding | Tekirdag, TR | 2024 – 2025

- Developed and built C#/.NET and Qt test GUIs for radar-simulator prototypes, enabling rapid operator feedback.
- Completed a personal project in RF antenna design (LPDA and Yagi-Uda arrays).

Key Achievements:

- Saved 135 hours/month with 99.5% accuracy by automating SAP materials tracking with UiPath REFramework.
- Received two awards: RPA Final Project Champion (2024) and “İyi Ki Bizimlesin” Performance Award (2025).

PROFESSIONAL EXPERIENCE

Intern / Trainee Engineer | Beko – Koç Holding | Tekirdag, TR | 2023 – 2024

- Enabled 40K hours/year automated tests by adding CAN and RS485 handlers to MediaTester.

Key Achievements:

- Received promotion after delivering MediaTester and PowerTester (Qt + Python) and saving 900 hours/month in QA time.
- Reduced field-update cycles by 90% by authoring Android fastboot OTA scripts.
- Recognized as Koç Group RPA Triathlon Winner (2023).
- Eliminated manual cable swapping during tests by developing automated USB/HDMI switching matrix control software.

IT Staff | MilMay Tekstil | Malatya, TR | 2023 – 2023

Core Responsibilities:

- Implemented PostgreSQL ISO 9001 document-control system and facilitated training for 50+ staff.

Key Achievements:

- Achieved zero non-conformities at the first ISO audit through paperless workflow.
- Reduced manual testing workload across multiple product lines by developing test automation solutions.

PROJECT EXPERIENCE

- Yagi-Uda Designer (2025) – Yagi-Uda antenna design software with DG7YBN/DK7ZB/DL6WU/WA5VJB presets and split/bent dipole options; runs full NEC-2 calculations (patterns/impedance/VSWR etc.) and exports NEC files.
- CoderAI (2025) – Reduced cycle-time by 80% using LLM-powered agent that turns natural-language specs to full repositories.
- TVPulse (2024) – Saved 1700 hours/month and executed 500+ scenarios nightly by automating an unattended IR regression framework.
- SAP Materials RPA (2024) – Saved 135 hours/month with an UiPath robot emailing daily stock demand.
- Breadboard 8-bit Computer (2022) – TTL-logic CPU running 6502 instruction subset at 10 MHz.
- TVBenchmark (2024) – Decreased manual research time by 90% using natural-language TV spec parser generating automated comparative benchmark reports.

EDUCATION

B.Sc., Computer Engineering
Inonu University / 2024
GPA: 3.38/4.0
Capstone: FPGA-based YOLO-v4 object detection at 18 fps on Cyclone V.

LANGUAGES

Turkish (Native) | English (C1 IELTS Academic 7.0)

REFERENCES

Aydoğan Bilir – Head of R&D Test & Verification, Beko Electronics Plant
Relationship: Direct manager (2023-2025)
Contact: aydogan.bilir@beko.com

Mustafa Bilgin – R&D Director, Beko Electronics Plant
Relationship: Skip-level manager and executive sponsor for my AI initiatives (2023-2025)
Contact: mustafa.bilgin@beko.com